

Stepper Motor Driver Project Proposal

Statement of Purpose

A stepper motor driver that receives a desired motor position over SPI and then executes a trajectory using trapezoidal motion profiling and closed loop PI control with feedback from an external quadrature encoder.

Pinout

Built In: clk, rst_n

Inputs: SPI (CLK, CS, MOSI), Encoder (A, B), Enable

Outputs: SPI MISO, Motor (STEP, DIR, ENABLE)*, Int

*Motor outputs drive an external IC that does power

Clock	
Frequency	10 MHz
SPI Peripheral	
SPI Mode	CPOL=0 / CPHA = 0 / MSB First
CLK Frequency	500 KHz
Frame Schema	1 2 bit address 13 bit payload 0 3 bit status (MISO)
Position & Motion	
Tick Storage	13 bit signed integer for target and current position
Control Frequency	1.2 KHz (10 MHz / 8192)
PI Coefficients	1, 1/2, 1/4, 1/8
Output	
PWM Width	8 bits
PWM Carrier	39 KHz (10 MHz / 256)
PWM Duty Cycle	1/256 to 255/256

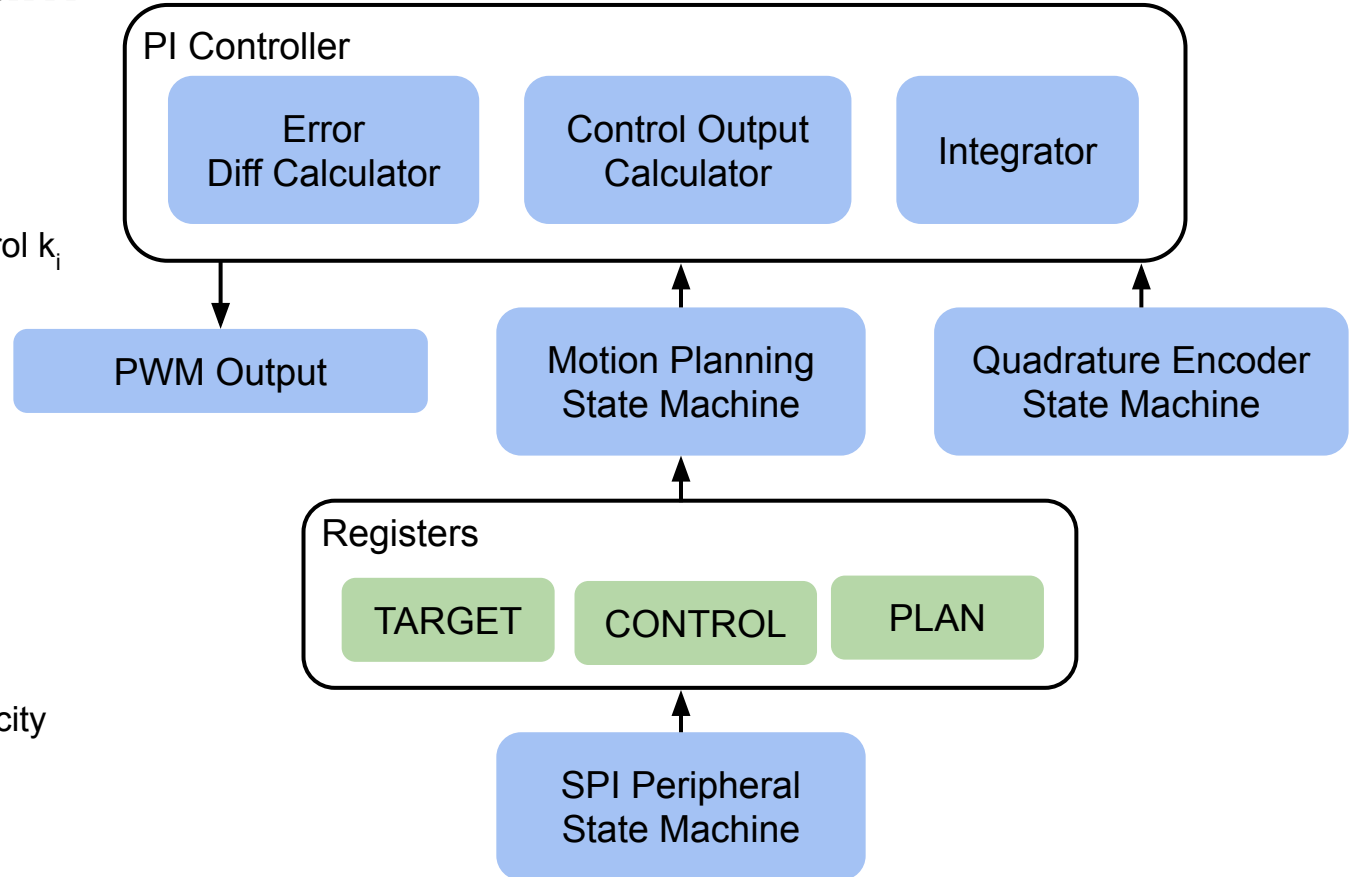
System Diagram

TARGET (0x0, 13 bits wide)
Target motor position.

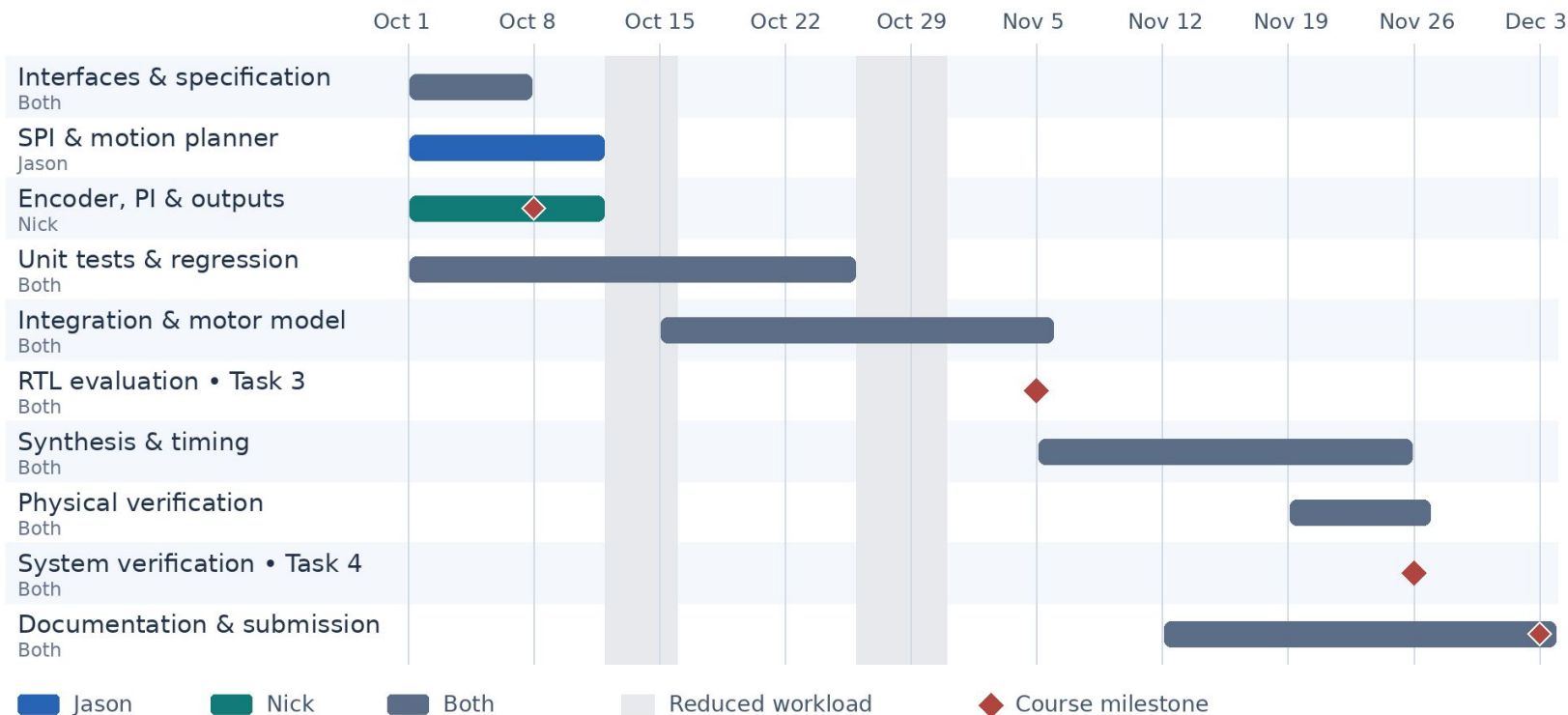
CONTROL (0x1, 4 bits wide)
2 MSB control k_p 2 LSB control k_i
0b11 1
0b10 $\frac{1}{2}$
0b01 $\frac{1}{4}$
0b00 $\frac{1}{8}$

PLAN (0x2, 6 bits wide)
3 MSB control maximum acceleration
0b000 100 steps/s²
0b111 10,000 steps/s²

3 LSB control maximum velocity
0b000 100 steps/s
0b111 10,000 steps/s



Timeline



Milestones: Oct 8 · Task 2 Nov 5 · Task 3 Nov 26 · Task 4 Dec 3 · Final submission

Shaded periods: reading week (Oct 12-15) and midterm week (Oct 26-30).